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Task Description-1

• Progressive Prompting for Calculator Design: Ask the AI to design a simple calculator program by initially providing only the function name. Gradually enhance the prompt by adding comments and usage examples.

Expected Output-1

• Comparison showing improvement in AI-generated calculator logic and structure.

Code:

AIAC_LAB 3.2.py

```python
def simple_calculator():
    print("Welcome to the Simple Calculator!")
    print("You can perform addition, subtraction, multiplication, and division.")

    while True:
        try:
            num1 = float(input("Enter the first number: "))
            operator = input("Enter the operator (+, -, *, /): ")
            num2 = float(input("Enter the second number: "))

            if operator == '+':
                result = num1 + num2
            elif operator == '-':
                result = num1 - num2
            elif operator == '*':
                result = num1 * num2
            elif operator == '/':
                if num2 != 0:
                    result = num1 / num2
                else:
                    print("Error: Division by zero is not allowed.")
                    continue
            else:
                print("Invalid operator. Please try again.")
                continue

            print(f"The result of {num1} {operator} {num2} is: {result}")

        except ValueError:
            print("Invalid input. Please enter numeric values.")

        cont = input("Do you want to perform another calculation? (yes/no): ")
        if cont.lower() != 'yes':
            print("Thank you for using the Simple Calculator. Goodbye!")
            break
if __name__ == "__main__":
    simple_calculator()
```

OutPut:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    SPELL CHECKER                    Python

PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 3.2.py"
Welcome to the Simple Calculator!
You can perform addition, subtraction, multiplication, and division.
Enter the first number: 9
Enter the operator (+, -, *, /): -
Enter the second number: 5
The result of 9.0 - 5.0 is: 4.0
Do you want to perform another calculation? (yes/no): no
Thank you for using the Simple Calculator. Goodbye!
PS C:\AI Assissted Coding Assessments>
```

Explanation :

In this we are desing a simple calculator . Where this perform a basic operation like addition ,subtraction,multiplication and division.

Task Description-2

• Refining Prompts for Sorting Logic: Start with a vague prompt for sorting student marks,

then refine it to clearly specify sorting order and constraints.

Expected Output-2

• AI-generated sorting function evolves from ambiguous logic to an accurate and efficient

implementation.

CODE:

```python
def sort_marks():
    n = len(marks)
    for i in range(n):
        for j in range(0, n-i-1):
            if marks[j] > marks[j+1]:
                marks[j], marks[j+1] = marks[j+1], marks[j]
# Example usage
marks = [85, 92, 78, 90, 88]
sort_marks()
print("Sorted marks:", marks)
```

OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    SPELL CHECKER

PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 3.2.py"
Sorted marks: [78, 85, 88, 90, 92]
PS C:\AI Assissted Coding Assessments>
```

EXPLANATION:

In this we are sorting the marks in ascending order.Where the number are given in random order but the output is in ascending order.

Task Description-3

• Few-Shot Prompting for Prime Number Validation: Provide multiple input-output examples for a function that checks whether a number is prime. Observe how few-shot prompting improves correctness.

Expected Output-3

• Improved prime-checking function with better edge-case handling.

CODE:

```python
def is_prime(n):
    if n <= 1:
        return False
    if n == 2:
        return True
    if n % 2 == 0:
        return False

    # check divisibility up to sqrt(n)
    i = 3
    while i * i <= n:
        if n % i == 0:
            return False
        i += 2
    return True

# Few-shot test cases
print(is_prime(2))   # True
print(is_prime(4))   # False
print(is_prime(17))  # True
print(is_prime(1))   # False
print(is_prime(-5))  # False
```

OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    SPELL CHECKER

PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 3.2.py"
Sorted marks: [78, 85, 88, 90, 92]
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 3.2.py"
True
False
True
False
False
PS C:\AI Assissted Coding Assessments>
```

EXPLANATION:

We are creating the function to check the number is prime or not with some sample test cases .

Task Description-4

• Prompt-Guided UI Design for Student Grading System: Create a user interface for a

student grading system that calculates total marks, percentage, and grade based on user

input.

Expected Output-4

• Well-structured UI code with accurate calculations and clear output display.

CODE:

```
def grading_system():
    print("=== Student Grading System ===")
    n = int(input("Enter number of subjects: "))

    marks = []
    for i in range(n):
        m = int(input(f"Enter marks for subject {i+1}: "))
        marks.append(m)

    total = sum(marks)
    percentage = total / n

    # Grade logic
    if percentage >= 90:
        grade = "A"
    elif percentage >= 75:
        grade = "B"
    elif percentage >= 50:
        grade = "C"
    else:
        grade = "F"

    print("\n--- Result ---")
    print(f"Total Marks: {total}")
    print(f"Percentage: {percentage:.2f}%")
    print(f"Grade: {grade}")
grading_system()
```

OUTPUT:

```
PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 3.2.py"
=== Student Grading System ===
Enter number of subjects: 5
Enter marks for subject 1: 50
Enter marks for subject 2: 53
Enter number of subjects: 5
Enter marks for subject 1: 50
Enter marks for subject 2: 53
Enter marks for subject 1: 50
Enter marks for subject 2: 53
Enter marks for subject 2: 53
Enter marks for subject 3: 2
Enter marks for subject 3: 2
Enter marks for subject 4: 5
Enter marks for subject 4: 5
Enter marks for subject 5: 55
Enter marks for subject 5: 55

--- Result ---
Total Marks: 165
Percentage: 33.00%
Grade: F
PS C:\AI Assissted Coding Assessments>
```

EXPLANATION:

In the code we are grading the student based on his marks and pass percentage and fail percentage and also the grade of the student.

Task Description-5

• Analyzing Prompt Specificity in Unit Conversion Functions: Improving a Unit Conversion Function (Kilometers to Miles and Miles to Kilometers) Using Clear Instructions.

Expected Output-5

• Analysis of code quality and accuracy differences across multiple prompt variations.

CODE:

```python
def convert_units(value, unit):
    if unit == "km":
        # kilometers to miles
        return round(value * 0.621371, 2)
    elif unit == "mi":
        # miles to kilometers
        return round(value / 0.621371, 2)
    else:
        return None  # invalid unit

# Few-shot examples
print(convert_units(10, "km"))  # 6.21 miles
print(convert_units(6.21, "mi"))  # 10.00 km
print(convert_units(0, "km"))  # 0.00 miles
print(convert_units(-5, "mi"))  # -8.05 km
```

OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    SPELL CHECKER

PS C:\AI Assissted Coding Assessments> & C:/Python314/python.exe "c:/AI Assissted Coding Assessments/AIAC_LAB 3.2.py"
6.21
9.99
0.0
-8.
Focus folder in explorer (ctrl + click)
PS C:\AI Assissted Coding Assessments>
```

EXPLANATION:

In the above we are convertin the units in cm,m,km,mi by using the function.